

Problem

A series-connected circuit has $R = 4\ \Omega$ and $L = 25\ \text{mH}$. (a) Calculate the value of C that will produce a quality factor of 50. (b) Find ω_1 , ω_2 , and B . (c) Determine the average power dissipated at $\omega = \omega_0$, ω_1 , ω_2 . Take $V_m = 100\ \text{V}$.

Step-by-step solution

Step 1 of 5

A Series – connected circuit has $R = 4\ \Omega$ and $L = 25\ \text{mH}$

a) Given that, $Q = 50$

We know that,

$$\text{Bandwidth } (B) = \frac{R}{L}$$

$$\Rightarrow B = \frac{4}{25 \times 10^{-3}}$$

$$\therefore B = 160\ \text{rad/sec}$$

We know that,

$$\text{Quality factor } (Q) = \frac{\omega_0}{B}$$

$$\Rightarrow 50 = \frac{\omega_0}{160}$$

$$\Rightarrow \omega_0 = 50 \times 160$$

$$\therefore \omega_0 = 8000\ \text{rad/sec}$$

Therefore, the resultant frequency $(\omega_0) = \frac{1}{\sqrt{LC}}$

$$\omega_0^2 = \frac{1}{LC}$$

$$C = \frac{1}{\omega_0^2 \times L}$$

Substitute values of ω_0 and L in the above equation,

$$C = \frac{1}{(8000)^2 \times (25 \times 10^{-3})}$$

$$\therefore \boxed{C = 0.625\ \mu\text{F}}$$

Step 2 of 5

b) We know that, the lower half – power frequency is,

$$\begin{aligned}\Rightarrow \omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \Rightarrow \omega_1 &= \frac{-4}{2 \times 25 \times 10^{-3}} + \sqrt{\left(\frac{4}{2 \times 25 \times 10^{-3}}\right)^2 + \frac{1}{(25 \times 10^{-3})(0.625 \times 10^{-6})}} \\ \Rightarrow \omega_1 &= \frac{-4}{0.05} + \sqrt{\left(\frac{4}{0.05}\right)^2 + (64 \times 10^6)} \\ \Rightarrow \omega_1 &= -80 + \sqrt{6400 + (64 \times 10^6)} \\ \Rightarrow \omega_1 &= -80 + \sqrt{64.0064 \times 10^6} \\ \Rightarrow \omega_1 &= -80 + (8 \times 10^3) \\ \therefore \omega_1 &= 7920 \text{ rad/sec}\end{aligned}$$

Step 3 of 5

The upper half – power frequency is

$$\begin{aligned}\omega_2 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \Rightarrow \omega_2 &= \frac{4}{2 \times (25 \times 10^{-3})} + \sqrt{\left(\frac{4}{2 \times (25 \times 10^{-3})}\right)^2 + \frac{1}{(25 \times 10^{-3})(0.625 \times 10^{-6})}} \\ \Rightarrow \omega_2 &= \frac{4}{0.05} + \sqrt{\left(\frac{4}{0.05}\right)^2 + (64 \times 10^6)} \\ \Rightarrow \omega_2 &= 80 + \sqrt{6400 + (64 \times 10^6)} \\ \Rightarrow \omega_2 &= 80 + \sqrt{64.0064 \times 10^6} \\ \Rightarrow \omega_2 &= 80 + (8 \times 10^3) \\ \therefore \omega_2 &= 8080 \text{ rad/sec}\end{aligned}$$

Thus, the Bandwidth (B) = $\frac{R}{L}$

$$\begin{aligned}\Rightarrow B &= \frac{4}{25 \times 10^{-3}} \\ \therefore B &= 160 \text{ rad/sec}\end{aligned}$$

Step 4 of 5

c) Given that,

$$V_m = 100 \text{ V}$$

The average power dissipated at resonant frequency (ω_0) is,

$$\begin{aligned}P(\omega_0) &= \frac{1}{2} \times \frac{V_m^2}{R} \\ \Rightarrow P(\omega_0) &= \frac{1}{2} \times \frac{(100)^2}{4} \\ \therefore P(\omega_0) &= 1.25 \text{ kW}\end{aligned}$$

At certain frequencies $\omega = \omega_1, \omega_2$ the dissipated power is,

$$P(\omega_1) = P(\omega_2) = \frac{V_m^2}{4R}$$

$$\Rightarrow P(\omega_1) = P(\omega_2) = \frac{(100)^2}{4 \times 4}$$

$$\Rightarrow P(\omega_1) = P(\omega_2) = 0.625 \text{ kW}$$

$$\therefore \boxed{P(\omega_1) = 0.625 \text{ kW}}$$

$$\boxed{P(\omega_2) = 0.625 \text{ kW}}$$